Smart Collectorate

Taking a District Magistrate's office from paper files to an AI-enabled workflow, and getting people to use it

Pulse Rajan · Applied AI Engineer assessment · Creditfix Group · November 2026

The office is busy, but no one can see the work

~4,000

grievances and applications a month

~30 days

average time to dispose of a grievance

1 in 3

citizens come back just to ask for status

0

live view of pending files for the DM

Why: paper files and registers, follow-ups on phone and WhatsApp groups, data in scattered spreadsheets, and no single owner for a case at each step.

Figures are working assumptions for a district of ~30 lakh people. They are measured properly in discovery and become the pilot baseline.

Discovery: meet the people who do the work

DM, ADM, SDMs

- Which three numbers do you want every Monday?
- Which delays cause the most complaints?

Clerks and inward desk

- Walk me through one file, receipt to disposal
- Where does your day go?

Field staff (Tehsildar, Patwari, BDO)

- How do reports reach you and go back?
- What phone, network and language do you use?

Citizens at the gate and helpdesk

- How many visits did this take?
- Would you file and track on WhatsApp?

2–3 weeks · 25–30 short interviews · 3 days sitting with the inward desk · no system is proposed until this is done

Mapping today's workflow to find the bottlenecks



Hand-written entry, no tracking number

File waits in a tray, no reminder

Report sent by phone or WhatsApp

Often never told; visits again

How we map it

- Trace 50 real files end to end and note the date at every hand-off
- Count hand-offs, waiting days and repeat visits for each step
- Rank steps by delay × volume; the top three become pilot targets
- Show the map back to the staff and correct it with them

Assumptions we make, and how we check them

Assumption	How we validate it
Most grievances arrive on paper or in person, mostly in Hindi	Count one month of the inward register by channel and language
About 10 categories cover 70% of all cases	Hand-tag 300 past grievances with the section clerks
Most staff have Android phones; the office has broadband	Quick device and network survey during interviews
Citizen data must stay in India on NIC or a government-approved cloud	Confirm with the District Informatics Officer (NIC)
The state grievance portal (CPGRAMS / IGRS) stays the system of record	Check API or data-export access with NIC
The DM will sponsor the pilot and name a nodal officer	Written go-ahead before the pilot starts

Seven use cases, ranked by impact

#	Use case	Why it matters	Type	Phase
1	Grievance intake and smart routing	One tracking number per case; right department on day one	AI	Pilot
2	File tracking with a QR code on every file	No lost files; anyone can see who holds a file now	Automation	Pilot
3	SLA reminders and escalation	Files stop waiting in trays without anyone knowing	Automation	Pilot
4	Live dashboard for the DM	Backlog and overdue cases by department, today	Automation	Pilot
5	WhatsApp helpline in Hindi and English	File and check status without visiting the office	AI	Rollout
6	Weekly review-meeting reports	Ready in minutes instead of two days of compiling	Automation + AI	Rollout
7	AI assistant for officers	Summarise a long file, draft a reply, find past orders	AI	Scale

Picked by: citizen pain × volume × how fast it can go live. The first four cover the biggest bottlenecks and need little change in how staff work.

Where AI is really needed, and where it isn't



Use an LLM: the input is messy language

- Reading free-text grievances in Hindi, English or a mix
- Suggesting department and urgency; pulling out place, scheme, names
- Summarising long files for an officer
- Drafting replies and meeting notes for an officer to approve
- Answering citizen questions on WhatsApp

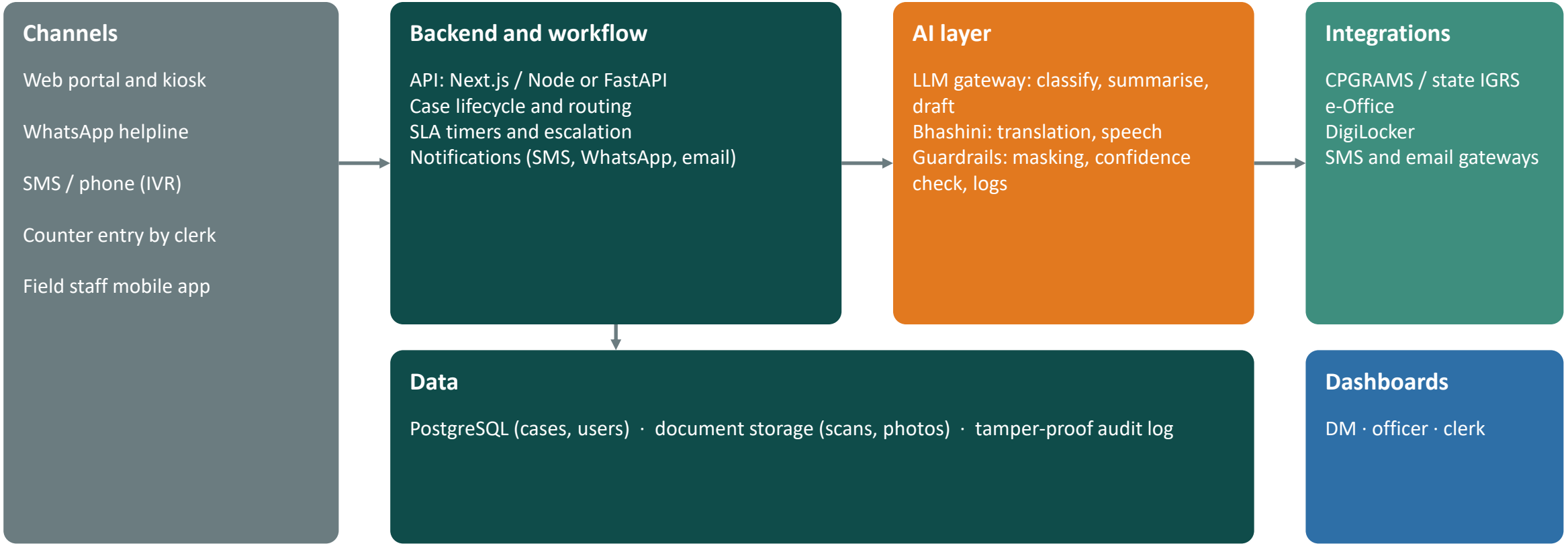


Use plain rules: the logic is clear

- Tracking numbers and QR codes on files
- SLA timers, reminders and escalation
- Dashboards and reports from data
- SMS and WhatsApp status updates
- Access control and the audit log

Our rule: if a simple rule can do the job, a rule does it. AI only suggests. A person makes the decision.

System architecture



Hosted in India on NIC or a MeitY-empanelled cloud · role-based login with OTP · every view and change logged · open-source, standard parts, so there is no lock-in

One grievance, start to finish



1. Citizen files

On WhatsApp, the portal or at the counter, in Hindi or English



2. Gets a number

Tracking ID by SMS straight away



3. AI suggests

Department, urgency, summary. Unsure? A clerk confirms



4. Officer acts

Sees the case, the summary and the deadline



5. Reminders

At 50% and 90% of the deadline; then escalated to the ADM



6. Closed

Citizen is told the outcome and rates the service

Citizens stop visiting for status updates. Officers see only their own queue. The DM sees everything.

Roadmap: zero to district-wide in 12 months



Discovery

Weeks 1–3

- Workflow map and baseline numbers
- Use cases signed off
- Hosting and data approval

Pilot

Weeks 4–14

- 2 departments, 1 tehsil
- Use cases 1–4 live
- 4-week parallel run with paper

Rollout

Months 4–6

- All departments, whole district
- WhatsApp helpline
- Automatic weekly reports

Scale

Months 7–12

- AI assistant for officers
- Field mobile app
- Portal integrations

Support

Ongoing

- Helpdesk and fixes
- Monthly AI accuracy review
- Training every quarter

Go / no-go gate at the end of the pilot (week 14): we only scale up if the pilot targets on slide 12 are met.

Team and effort for each phase

Phase	Team	Main deliverables	Effort
Discovery	Project lead / analyst, AI engineer, UX researcher	Workflow map, baseline, signed scope	~8 person-weeks
Pilot	Lead, 2 full-stack devs, AI engineer, QA, field trainer	Working system for 2 departments, trained staff	~45 person-weeks
Rollout	Pilot team + 1 developer, 2 trainers, DevOps (part-time)	District-wide system, WhatsApp helpline	~70 person-weeks
Scale	2 full-stack devs, AI engineer, QA, DevOps	Officer assistant, field app, integrations	~80 person-weeks
Support	Support engineer, helpdesk, AI engineer (part-time)	Uptime, fixes, model reviews, training	~1.5 people ongoing

On the office's side: one nodal officer (about 20% of their time) and one 'champion' clerk in each department.

What a successful pilot looks like

−40%

average time to dispose of a case

−30%

backlog older than 30 days

≥ 85%

AI routing accuracy (confirmed by officers)

≥ 80%

of officers using it every week

≥ 4 / 5

citizen rating of the service

0

lost files in pilot departments

All measured against the discovery baseline, for the same departments, after 10 weeks live.

Risks, and how we control them



Citizen data

- Follows the DPDP Act 2023: collect only what is needed, for a stated purpose
- Hosted in India on NIC or government-approved cloud
- Encrypted; role-based access; OTP login
- Every view and change recorded in an audit log
- Aadhaar and phone numbers masked before any AI call



AI mistakes

- AI suggests, an officer decides
- Below 70% confidence → goes to a person to check
- Fixed list of departments; AI cannot invent one
- Tested on 500 real Hindi/English cases; accuracy and bias reviewed monthly
- Original text always shown next to the AI summary



System and vendor

- If the AI is down, routing falls back to the clerk
- Daily backups; works on slow connections
- Security audit by a CERT-In empanelled auditor before go-live
- Open-source stack; data can be exported any time

Getting staff to actually use it

Built for the clerk

Hindi first, big buttons, fewer steps than the paper register

A champion in each department

One trusted clerk trained first, who helps colleagues

Short, hands-on training

30 minutes at their own desk, plus short Hindi videos

Paper stays for 4 weeks

Run both side by side until staff trust it

Help is one call away

Helpdesk number and a support WhatsApp group

The DM uses the dashboard

When the DM reviews from it in meetings, everyone keeps it updated

The pitch to the DM, in plain words

> Today, no one in this office can tell a citizen where their file is.
>
> In ten weeks, in two departments, every application will get a tracking number and reach the right desk on day one, and you will see every delayed file on one screen.
>
> Your staff still make every decision; the system reminds and suggests. Data stays in India, on government servers.
>
> If it does not cut delays by a third, you do not pay for the next phase.

About 60 seconds · no technical terms · ends with a low-risk offer

Handling the three common objections

“We already have a system”

We don't replace CPGRAMS, the state portal or e-Office. We fill the gaps between them (tracking, reminders, one dashboard) and send data back into them.

“It is not secure”

Hosted in India on government cloud, with OTP login, role-based access and a full audit log. Citizen data is never used to train AI. Security audit before go-live.

“Our staff will not use it”

It is in Hindi, with fewer steps than the register. Champions and desk-side training help. We track usage every week and fix whatever staff dislike.

Every answer ends with an offer to show it working in the pilot, not just a promise.

Engagement and pricing: pilot first, pay as it proves itself

Pilot

10–12 weeks

₹5–8 lakh

Fixed fee for 2 departments. Next phase only if the targets are met.

Rollout

Phased by module

₹20–30 lakh

Whole district, bought module by module through GeM.

Support

Yearly

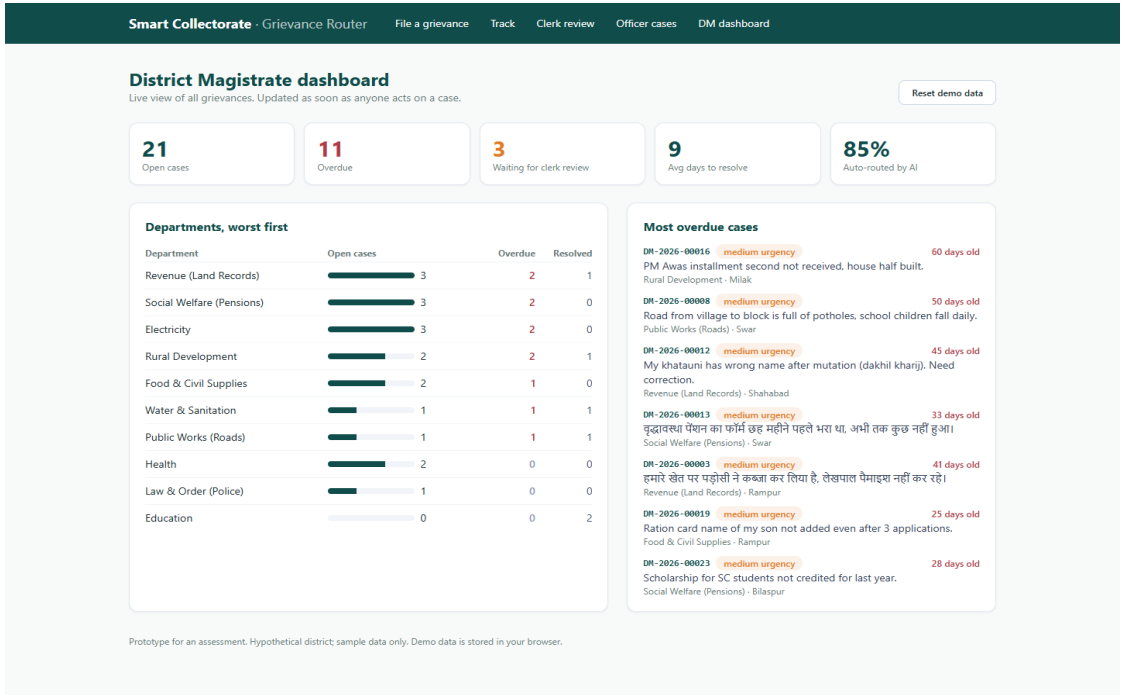
15–20% / yr

Of the build cost. Hosting, helpdesk, AI monitoring, updates and refresher training.

Indicative figures, to be firmed up after discovery. Discovery itself can be done at no cost to earn trust.

Working proof: AI grievance classifier and router

- Citizen types a complaint in Hindi, English or a mix
- Phone and Aadhaar numbers are masked before the AI sees the text
- AI suggests department, urgency and a one-line summary, with a confidence score
- Below 70% confidence, the case goes to the clerk's review queue
- Citizen gets a tracking ID; the deadline clock starts
- DM dashboard: backlog, overdue cases and load by department
- Works without an AI key (keyword fallback), so the demo never fails
- 20 test cases: AI (Claude Sonnet) 100% correct, keyword backup 90%; 0 sent to the wrong department



Next.js · TypeScript · any LLM (Claude / Gemini / OpenAI)

Code: GitHub link shared with the submission

Next steps

1. Approve a 3-week discovery (no cost)
2. Name a nodal officer and two pilot departments
3. Agree the pilot targets and the go / no-go date

Thank you · Questions welcome